

University Course Management System

Software Verification and Validation SRS



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1. Introduction

The University Course Management System is a software system developed to manage academic course related activities in a university environment. Universities manage a large amount of student and course information every semester. Handling this information manually can become difficult and time consuming. The system is developed to make these activities easier, faster, and more organized.

In many universities, students face problems during course registration because of manual work and poor record management. Students may register in wrong courses, miss prerequisite requirements, or face delays during enrollment. Administrators may also face difficulty in handling large numbers of students and courses at the same time. These problems reduce efficiency and may create confusion during the academic semester.

The University Course Management System helps solve these problems by providing an organized and automated platform. The system allows students to view courses, register in courses, and manage their enrollments. Administrators can create courses, update course details, define course capacities, and assign instructors. Instructors can view assigned courses and enrolled students.

One of the main goals of this system is to improve accuracy and consistency in academic management. Manual systems may create duplicate records, incorrect enrollment counts, or missing prerequisite checks. The proposed system follows predefined rules and conditions which help reduce such problems.

The system also focuses on maintaining proper academic discipline. Students cannot register in courses without completing required prerequisites. Students are also not allowed to enroll in courses that are already full. These conditions help maintain fairness and proper academic flow.

This project is developed as part of the Software Verification and Validation Lab. The main focus of the project is not only software design but also formal verification and validation. Formal verification helps ensure that the system behaves correctly under different conditions and constraints.

Formal methods help detect logical errors and inconsistencies before system implementation. In this project, different formal verification techniques are applied to verify system correctness. Z notation is used for formal modeling, VDM is used for functional specification, and Alloy Analyzer is used for structural verification and counterexample analysis.

Verification is important because software systems must behave correctly in all possible situations. If academic systems fail, students may lose registrations or academic records

may become inconsistent. Formal verification reduces these risks by mathematically verifying system operations and constraints.

The project also includes validation and security activities. Validation ensures that the system requirements are satisfied correctly. Security analysis helps identify possible weaknesses and vulnerabilities. GitHub Actions and OWASP ZAP are used as part of the validation process.

The University Course Management System was selected because it contains clear workflows and academic constraints that are suitable for formal verification. Operations such as course enrollment, prerequisite validation, and instructor assignment can be modeled properly using formal methods.

The system includes three main actors which are students, instructors, and administrators. Each actor interacts with the system according to defined permissions and responsibilities. This creates a structured and organized academic environment.

Overall, the University Course Management System provides a reliable solution for managing university course activities. The system improves efficiency, reduces manual work, maintains consistency, and supports verification and validation activities.

2. Purpose

1. To define all requirements of the University Course Management System clearly.

This objective is important because it helps maintain proper verification and academic management during system development.

2. To explain how the system should behave during different academic operations.

This objective is important because it helps maintain proper verification and academic management during system development.

3. To provide proper documentation for the Software Verification and Validation project.

This objective is important because it helps maintain proper verification and academic management during system development.

4. To support formal modeling and verification activities.

This objective is important because it helps maintain proper verification and academic management during system development.

5. To reduce misunderstanding between developers and users.

This objective is important because it helps maintain proper verification and academic management during system development.

6. To provide a foundation for Z notation specifications.

This objective is important because it helps maintain proper verification and academic management during system development.

7. To support VDM based functional verification.

This objective is important because it helps maintain proper verification and academic management during system development.

8. To support Alloy based structural verification.

This objective is important because it helps maintain proper verification and academic management during system development.

9. To improve correctness and consistency of the proposed system.

This objective is important because it helps maintain proper verification and academic management during system development.

10. To identify possible requirement defects before implementation.

This objective is important because it helps maintain proper verification and academic management during system development.

11. To maintain proper academic workflows inside the system.

This objective is important because it helps maintain proper verification and academic management during system development.

12. To help developers understand the expected system behavior.

This objective is important because it helps maintain proper verification and academic management during system development.

13. To improve the quality and reliability of the software system.

This objective is important because it helps maintain proper verification and academic management during system development.

14. To support validation and testing activities during development.

This objective is important because it helps maintain proper verification and academic management during system development.

3. Scope

The University Course Management System focuses on managing academic course related activities in a university environment. The system includes course management, student enrollment, instructor assignment, and prerequisite validation.

The system allows administrators to create new courses and update existing course information. Course details include course ID, course title, credit hours, prerequisites, and maximum capacity.

The system allows students to register in available courses during the semester. Students can view course information before enrollment.

The system verifies prerequisite requirements before enrollment confirmation. Students cannot enroll in advanced courses without completing required previous subjects.

The system manages course capacities and prevents over enrollment. If a course becomes full, additional enrollment requests are rejected automatically.

The system allows students to drop courses during the allowed registration period. Enrollment counts are updated automatically after course removal.

The system allows instructors to view assigned courses and enrolled student lists. This helps instructors maintain proper awareness of class registrations.

The system maintains unique student IDs and course IDs to avoid duplicate records and confusion.

The system focuses on correctness and consistency of academic records during all operations.

The project does not include online fee payment services because financial management is outside the scope of this system.

The project does not include attendance management or biometric attendance tracking.

The project does not include examination management, grading systems, or assignment submission features.

The project does not include mobile application development because the focus is on formal verification and validation.

The project also excludes advanced analytics and artificial intelligence features.

4. System Overview

The University Course Management System works with three main users which are students, instructors, and administrators. Each user performs different tasks inside the system.

Students use the system to view courses and register for subjects offered during the semester. Before enrollment confirmation, the system checks course prerequisites and available seats.

If the student satisfies all conditions, the enrollment request is accepted. Otherwise, the system rejects the request and displays an appropriate message.

Students can also drop registered courses during the allowed period. The system automatically updates course records after a course is dropped.

Instructors use the system to view assigned courses and lists of enrolled students. This helps instructors maintain awareness of academic activities related to their courses.

Administrators have the highest level of access inside the system. They manage courses, define prerequisites, assign instructors, and monitor enrollment activities.

The workflow of the system starts when administrators create semester course offerings. After that, students log into the system and begin course registration.

The system validates every operation according to predefined rules and constraints. Invalid operations are automatically rejected.

The system maintains enrollment consistency and prevents invalid states such as duplicate registration or exceeding course capacity.

The system stores student information, course information, and enrollment records in an organized way. This helps improve academic management and reduces manual work.

The University Course Management System focuses on reliability, consistency, and correctness during every academic operation.

5. Objectives

1. To automate university course registration processes.

This objective helps improve the quality and reliability of the proposed system.

2. To reduce manual work during academic management.

This objective helps improve the quality and reliability of the proposed system.

3. To maintain accurate enrollment records.

This objective helps improve the quality and reliability of the proposed system.

4. To ensure prerequisite validation during registration.

This objective helps improve the quality and reliability of the proposed system.

5. To maintain course capacity limits.

This objective helps improve the quality and reliability of the proposed system.

6. To improve reliability of course management activities.

This objective helps improve the quality and reliability of the proposed system.

7. To support secure and organized academic operations.

This objective helps improve the quality and reliability of the proposed system.

8. To verify system correctness using formal methods.

This objective helps improve the quality and reliability of the proposed system.

9. To apply Z notation for formal modeling.

This objective helps improve the quality and reliability of the proposed system.

10. To apply VDM for functional specification.

This objective helps improve the quality and reliability of the proposed system.

11. To apply Alloy Analyzer for structural verification.

This objective helps improve the quality and reliability of the proposed system.

12. To perform validation and security analysis.

This objective helps improve the quality and reliability of the proposed system.

13. To improve understanding of Software Verification and Validation concepts.

This objective helps improve the quality and reliability of the proposed system.

14. To maintain consistency during enrollment operations.

This objective helps improve the quality and reliability of the proposed system.

15. To provide a structured academic management environment.

This objective helps improve the quality and reliability of the proposed system.

6. Functional Requirements

FR1

The system shall be able to allow administrators to add new courses.

Description: This requirement supports proper academic management and system consistency.

Priority: High

Input: User request or academic information.

Output: Correct academic operation performed by the system.

Verification Method: Formal verification and validation activities.

Expected Result: The system performs the operation correctly according to academic rules.

FR2

The system shall be able to update existing course details.

Description: This requirement supports proper academic management and system consistency.

Priority: High

Input: User request or academic information.

Output: Correct academic operation performed by the system.

Verification Method: Formal verification and validation activities.

Expected Result: The system performs the operation correctly according to academic rules.

FR3

The system shall be able to remove inactive courses from the system.

Description: This requirement supports proper academic management and system consistency.

Priority: High

Input: User request or academic information.

Output: Correct academic operation performed by the system.

Verification Method: Formal verification and validation activities.

Expected Result: The system performs the operation correctly according to academic rules.

FR4

The system shall be able to allow students to view available courses.

Description: This requirement supports proper academic management and system consistency.

Priority: High

Input: User request or academic information.

Output: Correct academic operation performed by the system.

Verification Method: Formal verification and validation activities.

Expected Result: The system performs the operation correctly according to academic rules.

FR5

The system shall be able to display course prerequisites to students.

Description: This requirement supports proper academic management and system consistency.

Priority: High

Input: User request or academic information.

Output: Correct academic operation performed by the system.

Verification Method: Formal verification and validation activities.

Expected Result: The system performs the operation correctly according to academic rules.

FR6

The system shall be able to allow students to enroll in available courses.

Description: This requirement supports proper academic management and system consistency.

Priority: High

Input: User request or academic information.

Output: Correct academic operation performed by the system.

Verification Method: Formal verification and validation activities.

Expected Result: The system performs the operation correctly according to academic rules.

FR7

The system shall be able to verify prerequisites before enrollment.

Description: This requirement supports proper academic management and system consistency.

Priority: High

Input: User request or academic information.

Output: Correct academic operation performed by the system.

Verification Method: Formal verification and validation activities.

Expected Result: The system performs the operation correctly according to academic rules.

FR8

The system shall be able to reject enrollment when prerequisites are incomplete.

Description: This requirement supports proper academic management and system consistency.

Priority: High

Input: User request or academic information.

Output: Correct academic operation performed by the system.

Verification Method: Formal verification and validation activities.

Expected Result: The system performs the operation correctly according to academic rules.

FR9

The system shall be able to prevent duplicate course enrollment.

Description: This requirement supports proper academic management and system consistency.

Priority: High

Input: User request or academic information.

Output: Correct academic operation performed by the system.

Verification Method: Formal verification and validation activities.

Expected Result: The system performs the operation correctly according to academic rules.

FR10

The system shall be able to prevent enrollment when course capacity becomes full.

Description: This requirement supports proper academic management and system consistency.

Priority: High

Input: User request or academic information.

Output: Correct academic operation performed by the system.

Verification Method: Formal verification and validation activities.

Expected Result: The system performs the operation correctly according to academic rules.

FR11

The system shall be able to allow students to drop registered courses.

Description: This requirement supports proper academic management and system consistency.

Priority: High

Input: User request or academic information.

Output: Correct academic operation performed by the system.

Verification Method: Formal verification and validation activities.

Expected Result: The system performs the operation correctly according to academic rules.

FR12

The system shall be able to update enrollment counts automatically.

Description: This requirement supports proper academic management and system consistency.

Priority: High

Input: User request or academic information.

Output: Correct academic operation performed by the system.

Verification Method: Formal verification and validation activities.

Expected Result: The system performs the operation correctly according to academic rules.

FR13

The system shall be able to assign instructors to courses.

Description: This requirement supports proper academic management and system consistency.

Priority: High

Input: User request or academic information.

Output: Correct academic operation performed by the system.

Verification Method: Formal verification and validation activities.

Expected Result: The system performs the operation correctly according to academic rules.

FR14

The system shall be able to maintain unique student IDs.

Description: This requirement supports proper academic management and system consistency.

Priority: High

Input: User request or academic information.

Output: Correct academic operation performed by the system.

Verification Method: Formal verification and validation activities.

Expected Result: The system performs the operation correctly according to academic rules.

FR15

The system shall be able to maintain unique course IDs.

Description: This requirement supports proper academic management and system consistency.

Priority: High

Input: User request or academic information.

Output: Correct academic operation performed by the system.

Verification Method: Formal verification and validation activities.

Expected Result: The system performs the operation correctly according to academic rules.

FR16

The system shall be able to store student registration records.

Description: This requirement supports proper academic management and system consistency.

Priority: High

Input: User request or academic information.

Output: Correct academic operation performed by the system.

Verification Method: Formal verification and validation activities.

Expected Result: The system performs the operation correctly according to academic rules.

FR17

The system shall be able to display enrollment confirmation messages.

Description: This requirement supports proper academic management and system consistency.

Priority: High

Input: User request or academic information.

Output: Correct academic operation performed by the system.

Verification Method: Formal verification and validation activities.

Expected Result: The system performs the operation correctly according to academic rules.

FR18

The system shall be able to reject invalid enrollment requests.

Description: This requirement supports proper academic management and system consistency.

Priority: High

Input: User request or academic information.

Output: Correct academic operation performed by the system.

Verification Method: Formal verification and validation activities.

Expected Result: The system performs the operation correctly according to academic rules.

FR19

The system shall be able to allow administrators to monitor enrollment activities.

Description: This requirement supports proper academic management and system consistency.

Priority: High

Input: User request or academic information.

Output: Correct academic operation performed by the system.

Verification Method: Formal verification and validation activities.

Expected Result: The system performs the operation correctly according to academic rules.

FR20

The system shall be able to maintain course information consistently.

Description: This requirement supports proper academic management and system consistency.

Priority: High

Input: User request or academic information.

Output: Correct academic operation performed by the system.

Verification Method: Formal verification and validation activities.

Expected Result: The system performs the operation correctly according to academic rules.

7. Non Functional Requirements

NFR1

The system shall be able to provide secure access to authorized users only.

This requirement improves the quality and reliability of the system.

NFR2

The system shall be able to maintain consistent academic records.

This requirement improves the quality and reliability of the system.

NFR3

The system shall be able to support multiple users at the same time.

This requirement improves the quality and reliability of the system.

NFR4

The system shall be able to respond to user requests quickly.

This requirement improves the quality and reliability of the system.

NFR5

The system shall be able to provide reliable enrollment services.

This requirement improves the quality and reliability of the system.

NFR6

The system shall be able to protect student information from unauthorized access.

This requirement improves the quality and reliability of the system.

NFR7

The system shall be able to operate continuously during registration periods.

This requirement improves the quality and reliability of the system.

NFR8

The system shall be able to maintain data accuracy during all operations.

This requirement improves the quality and reliability of the system.

NFR9

The system shall be able to provide user friendly interfaces.

This requirement improves the quality and reliability of the system.

NFR10

The system shall be able to support future scalability.

This requirement improves the quality and reliability of the system.

8. System Constraints

Course enrollment cannot exceed maximum course capacity.

This constraint helps maintain system correctness and academic consistency.

Students cannot register in the same course more than once.

This constraint helps maintain system correctness and academic consistency.

Students must complete prerequisites before advanced course enrollment.

This constraint helps maintain system correctness and academic consistency.

Only administrators can add or update courses.

This constraint helps maintain system correctness and academic consistency.

Only authorized instructors can teach courses.

This constraint helps maintain system correctness and academic consistency.

Course IDs must remain unique.

This constraint helps maintain system correctness and academic consistency.

Student IDs must remain unique.

This constraint helps maintain system correctness and academic consistency.

The system depends on internet availability.

This constraint helps maintain system correctness and academic consistency.

The university server must remain operational during registration periods.

This constraint helps maintain system correctness and academic consistency.

The system must follow university academic policies.

This constraint helps maintain system correctness and academic consistency.

9. Assumptions

All students have valid university accounts.

This assumption is necessary for proper system operation and management.

All instructors have valid login credentials.

This assumption is necessary for proper system operation and management.

Users have proper internet access.

This assumption is necessary for proper system operation and management.

Administrators enter correct academic information.

This assumption is necessary for proper system operation and management.

Students understand prerequisite requirements.

This assumption is necessary for proper system operation and management.

The university provides stable server support.

This assumption is necessary for proper system operation and management.

Users follow university policies during system usage.

This assumption is necessary for proper system operation and management.

Students provide correct personal information.

This assumption is necessary for proper system operation and management.

Instructors manage assigned courses properly.

This assumption is necessary for proper system operation and management.

The university maintains backup records for safety.

This assumption is necessary for proper system operation and management.

10. Requirement Defect Taxonomy Table

Requirement	Defect Type	Reason
System should be secure	Vagueness	Security level is not defined
Fast registration process	Ambiguity	Fast is not measurable
Reliable academic system	Non Verifiable	Reliability criteria missing
Efficient enrollment	Vagueness	Efficiency is unclear
Simple interface	Ambiguity	Simple is subjective
Good performance	Non Verifiable	Performance limits missing
Flexible management	Vagueness	Flexibility is undefined
Quick response	Ambiguity	Quick is not measurable

The requirement defect taxonomy table helps identify unclear and incomplete requirements before implementation.

Requirement defects may create misunderstanding and verification problems during software development.

Identifying defects early helps improve software quality and correctness.

11. GitHub Issues Log

Issue 1

Requirement NFR1 contains vague security description.

Label: Vagueness

This issue was identified during requirement analysis and documented for verification purposes.

Issue 2

Enrollment response time is not measurable.

Label: Ambiguity

This issue was identified during requirement analysis and documented for verification purposes.

Issue 3

System reliability requirement is incomplete.

Label: Non Verifiable

This issue was identified during requirement analysis and documented for verification purposes.

Issue 4

Course capacity rules need clearer constraints.

Label: Incomplete

This issue was identified during requirement analysis and documented for verification purposes.

Issue 5

Performance requirements need exact values.

Label: Ambiguity

This issue was identified during requirement analysis and documented for verification purposes.

Issue 6

User interface requirement is subjective.

Label: Vagueness

This issue was identified during requirement analysis and documented for verification purposes.

12. Conclusion

This Software Requirements Specification document defines the requirements of the University Course Management System clearly and properly.

The document explains system behavior, user interactions, academic workflows, and operational constraints.

The requirements defined in this document provide a strong foundation for formal verification and validation activities.

The SRS supports formal modeling using Z notation, VDM specifications, and Alloy Analyzer.

The project focuses on correctness, consistency, reliability, and validation of academic operations.

Overall, the document helps improve software quality and supports proper academic management through verification and validation techniques.

Additional Discussion on Academic Management

Academic management is an important part of every university environment. Universities handle many students, instructors, and courses every semester. Managing this information manually can create delays and errors. The University Course Management System helps improve this process by providing a centralized platform for handling course related activities.

The system improves communication between students, instructors, and administrators. Students can easily view courses and manage registrations. Administrators can manage course records more efficiently. Instructors can stay informed about enrolled students and assigned subjects.

Another important benefit of the system is consistency. Academic records remain updated automatically after every operation. This reduces chances of missing information and duplicate records.

The proposed system also improves transparency in academic operations. Students can clearly see course prerequisites, capacities, and registration conditions before enrollment.

Importance of Verification and Validation

Software verification and validation are very important for systems that manage academic information. Incorrect enrollment records or missing prerequisite checks may create serious academic problems for students and universities.

Verification helps ensure that the system behaves according to defined requirements and constraints. Validation helps confirm that the system satisfies user needs and expectations.

Formal verification methods reduce logical errors by mathematically checking operations and system states. This increases reliability and confidence in the software system.

The University Course Management System is suitable for verification activities because it contains clear rules and workflows that can be modeled formally.

Benefits of Automated Academic Systems

Automated systems reduce the amount of manual work required during registration periods. This helps universities save time and improve efficiency.

Students can register in courses without visiting university offices physically. This improves convenience and reduces registration delays.

Administrators can manage a large amount of academic information more effectively through automated record keeping and validation checks.

The system also helps maintain fairness because all students follow the same registration rules and constraints.

Appendix A: Sample Course Registration Scenario

A student logs into the University Course Management System using valid university credentials. After successful login, the student views all available courses offered during the semester.

The student selects a course for enrollment. Before confirming registration, the system checks whether the student has completed the required prerequisite courses.

The system also checks whether seats are available in the selected course. If the course capacity is full, the enrollment request is rejected automatically.

If all conditions are satisfied, the system enrolls the student successfully and updates enrollment records immediately.

The instructor assigned to the course can later view the updated list of enrolled students through the system.

Appendix B: Academic Rules Supported by the System

Students cannot register in the same course multiple times.

Students must complete prerequisite courses before advanced course enrollment.

Course capacities cannot exceed predefined limits.

Only authorized administrators can update course information.

Only valid instructors can be assigned to university courses.

Course records must remain consistent during all operations.

Enrollment counts must update automatically after registration or course drop operations.

Appendix C: Future Improvements

Future versions of the system may include online examination management.

Attendance tracking and grading systems may also be integrated in the future.

The system may later support mobile applications for easier student access.

Advanced reporting and analytics features may be added for university administration.

Future versions may also support online payment systems and automated notifications.